[INCH-POUND]
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SUPERSEDING
MIL-C-80021B
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COMMERCIAL ITEM DESCRIPTION

CUTOFF MACHINE, ABRASIVE DISK, CHOP-STROKE

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-C-80021B and MIL-C-80071B for all federal agencies.

1. SCOPE. This commercial item description (CID) covers chop-stroke, abrasive disk cutoff machines intended for use in cutting materials such as hardened or annealed ferrous metals, nonferrous metals, plastics, ceramics, and refractory materials in the form of solid bars, tubing, structural sections, and extrusions. The machine is not intended to cut wood or wood products. A circular saw blade may be specified for use in cutting nonferrous materials other than wood or wood products.

2. CLASSIFICATION. The cutoff machines shall be of the following types, classes, styles, and sizes. The type, class, style, and size to be furnished shall be as specified (see 7.2(b)). Table I shows cutoff machine sizes and availability.

Type I - Dry cut
Type II - Wet cut

Class 1 - Manual-operated wheelhead
Class 2 - Power-operated wheelhead

Style A - Floor mounted
Style B - Bench mounted

Size – Sizes are shown in table I

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3405

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TABLE I. Cutoff machine sizes.

Size	Disk diameter, inches	Spindle drive motor, horsepower	Class 1		Class 2
			Style A	Style B	Style A
08	8	3	X	X	
10	10	3	X	X	X
12	12	5	X	X	X
14	14	5	X	X	X
16	16	7.5	X	X	X
20	20	15	X		X
24	24	25	X		X
26	26	30	X		X
30	30	30	X		X

3. SALIENT CHARACTERISTICS

3.1 General requirements. The machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. The design and construction of the machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2 Components. The cutoff machine shall include, as a minimum, the following components:

3.2.1 Base. The base of the cutoff machine shall support the workpiece holder and wheelhead and incorporate materials, shape, and mass to minimize adverse vibration and deflection. The base of the style A machine shall provide a means for anchoring the machine to the floor. The base of the style B machine shall provide a means for anchoring the machine to a bench or stand.

3.2.2 Workpiece holder. Unless otherwise specified (see 7.2(c)), the cutoff machine shall have a fixed workpiece holder. The workpiece holder shall be a vise, fence, V-block, or L-block as specified (see 7.2(d)). The workpiece holder for a style A cutoff machine shall be operated by hand, foot, or accessory control as specified (see 7.2(e)). The workpiece holder for a style B cutoff machine shall be hand controlled. If an adjustable workpiece holder is provided, it shall be capable of swiveling through one of the following horizontal arcs as specified (see7.2(f)):

a. Left 45 degrees.
b. Right 45 degrees.
c. Right 45 degrees and left 45 degrees.

3.2.3 Wheelhead. The wheelhead shall consist of a wheelhead arm, spindle, spindle drive system, wheel guards, and spindle drive motor for the abrasive disk or saw blade. A means shall be provided to prevent the flanges holding the abrasive disk from contacting the workpiece during a cutting cycle. Unless otherwise specified (see 7.2(g)), the cutoff machine shall have a

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nonswiveling wheelhead. A counterbalance system shall be provided on a class 1 machine to return the wheelhead to the normal up position upon release of manual down force. A power-driven mechanism shall be provided on a class 2 machine to move the wheelhead through a complete cutting cycle. The power-driven mechanism shall be actuated by hand, foot, or accessory control as specified (see 7.2(h)). If a swiveling wheelhead is provided, it shall be capable of swiveling through one of the following horizontal arcs as specified (see 7.2(i)):

a. Left 45 degrees.
b. Right 45 degrees.
c. Right 45 degrees and left 45 degrees.

3.2.3.1 Wheelhead arm. The wheelhead arm pivot bearings shall be constructed to maintain bearing lubrication and prevent the entrance of coolant, cuttings, abrasive dust, or other materials.

3.2.3.2 Spindle drive system. Unless otherwise specified (see 7.2(j)), the spindle motor horsepower shall be as listed in table I. The spindle drive system shall transfer full power of the motor to the spindle without slippage. If a belt system is used, the system shall have a means for adjusting and maintaining proper tension of the drive belts.

3.2.3.3 Wheel guards. Wheel guards for the abrasive disk shall not interfere with the operation of the machine and shall be removable to facilitate cleaning and servicing the machine.

3.2.3.4 Abrasive disk. Unless otherwise specified (see 7.2(k)), the abrasive disk provided shall be that considered by the manufacturer as standard.

3.2.3.5 Saw blade. If required, a circular saw blade shall be provided as specified (see 7.2(l)).

3.2.3.6 Flanges. Unless otherwise specified (see 7.2(m)), flanges considered by the manufacturer as standard shall be provided to attach the abrasive disk and circular saw blade to the spindle.

3.2.4 Oscillating (reciprocating) mechanism. When specified (see 7.2(n)), a mechanism shall be provided to oscillate (reciprocate) the spindle and abrasive disk in a direction parallel to the kerf in the workpiece.

3.2.5 Controls. All controls shall be located convenient to the operator.

3.2.6 Coolant system. The type II cutoff machine shall have a recirculating coolant system consisting of a sump or reservoir; a power-driven pump; a means for collecting spent coolant; and all piping, channeling, or ducting necessary to direct the coolant to the abrasive disk and workpiece and return it to the sump or reservoir. The reservoir capacity and pump flow shall be adequate to dissipate heat generated by the cutting process while cutting at the maximum cutting rate.

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3.2.7 Hydraulic system. When provided, the hydraulic system shall be sufficient for maximum loads and shall be complete including all pumps, piping, cylinders, and pressure controls. Overpressure protection shall be provided to prevent damage to components. The hydraulic reservoir shall have a means to determine the fluid level.

3.2.8 Pneumatic system. When provided, the pneumatic system shall be capable of operating on a 90- to 110-pounds-per-square-inch shop air-supply system. The pneumatic system shall consist of an air filter, pressure regulator, necessary gauges, operating valves, operating cylinders, relief valves, piping, hoses, and fittings.

3.2.9 Electrical system. The electrical system shall conform to the American National Standards Institute/National Fire Protection Association (ANSI/NFPA) 79, "Machinery, Industrial, Electrical Standard For". The machine shall be wired for a single 230-volt, 3-phase, 60-hertz circuit or a single 460-volt, 3-phase, 60-hertz circuit as specified (see 7.2(o)). If required, termination of wiring for connection to the power source shall be as specified (see 7.2(p)).

3.2.10 Electric motors. The drive motor(s) shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG-1, "Motors and Generators". Unless otherwise specified (see 7.2(q)), the spindle drive motor shall correspond to a minimum 30-minute duty cycle rating at the horsepower shown in table I. All nonspindle drive motors shall correspond to a minimum 30-minute duty cycle rating and shall be sized and rated not to limit the operation of the machine. Unless otherwise specified (see 7.2(r)), all motors shall be totally enclosed fan-cooled (TEFC).

3.2.11 Optional accessories. Optional accessories shall be provided as specified (see 7.2(s)).

3.3 Performance. Unless otherwise specified (see 7.2(t)), the cutoff machine shall meet the performance requirements and capacities shown in tables II and III, respectively, when tested in accordance with sections 5.3 through 5.4.

TABLE II. Performance requirements.

Test	Type I		Type II
	Size12 and under	Size 14 and over	All sizes
Minimum cutting rate, square inches per minute	10	15	2
Accuracy of 90-degree cut, inch per inch of diameter	±0.005	±0.005	±0.010
Surface finish, maximum average roughness, microinches	70	70	40

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TABLE III. Cutoff machine capacities.

Size	Capacity	
	Ferrous round bar, inches	Ferrous round tube, inches
08	1	1.5
10	1	2
12	1.75	2
14	2	3
16	2	3
20	3	5
24	6	6
26	5	6
30	8	8

3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(u)).

3.5 Safety and health requirements. The machine shall be designed and manufactured in accordance with ANSI B11.9, "Machines, Grinding, Safety Requirements for the Construction, Care and Use Of". When a circular saw blade is provided, the cutoff machine shall also be designed and manufactured in accordance with ANSI B11.10, "Sawing Machines, Metal, Safety Requirements for the Construction, Care and Use Of". The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910, "Occupational Safety and Health Standards for General Industry". If a conflict arises between the ANSI and OSHA standards, the OSHA standards shall apply.

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(v)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(w)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.7 Lubrication plate or chart. When specified (see 7.2(x)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application
b. Servicing interval
c. Type of lubricant(s) with SAE number or lubricant identifier

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3.8 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(y)), the nameplate shall contain the following information:

a. Nomenclature
b. Manufacturer's name
c. Serial number
d. Machine model designation
e. Power input (volts, total amperes, phase, frequency)
f. Short-circuit/over-current rating
g. Contract number or order number
h. National stock number
i. Date of manufacture

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The cutoff machine shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(z)), the cutoff machine shall be tested in accordance with sections 5.3.1 through 5.3.7.

5.3.1 Test setup and conditions. Unless otherwise specified (see 7.2(aa)), the manufacturer shall be responsible for supplying all tooling and material(s) and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.2(bb)).

5.3.2 Operational test. The cutoff machine shall be operated in accordance with the manufacturer's standard operating procedures. The cutoff machine shall be operated for not less

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than 30 minutes. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Performance tests. The performance tests shall be preformed with an SAE 1020 solid round steel bar having a nominal diameter equal to the maximum rated capacity shown in table III for the size of the machine.

5.3.4 Cutting rate test. All cuts shall be at a 90-degree angle. The cutoff machine shall cut four pieces not less than 1 inch thick from the workpiece. There shall be no evidence of damage (wear excepted) to the abrasive disk. The actual cutting time shall be recorded and the average cutting rate calculated. The cutting rate shall meet the requirements shown in table II.

5.3.5 Accuracy test. The last piece cut in the cutting rate test (see 5.3.4) shall be at least 4 inches long. The squareness and the surface finish of the cut faces shall meet the performance requirements shown in table II.

5.3.6 Angular cuts. Machines with a swivel workpiece holder shall set the holder at the maximum angle and one piece shall be cut. The cut face shall be within ±1/2 degree of the desired angle. Machines provided with a swivel wheelhead shall set the wheelhead at the maximum angle and one piece shall be cut. The cut face shall be within ±1/2 degree of the desired angle.

5.3.7 Optional tests. Optional tests shall be conducted as specified (see 7.2(cc)).

5.4 Acceptance. Unless otherwise specified (see 7.2(dd)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. Failure of the machine to meet the performance requirements and cutting capacities shown in tables II and III, respectively, shall be cause for rejection.

6. PACKAGING

6.1 Preservation, packing, and marking. For acquisition purposes, the products shall be preserved, packed, and marked as specified (see 7.2(ee)).

7. NOTES

7.1 Sources of documents.

7.1.1 FAR. The FAR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.

7.1.2 ANSI standards. Copies of ANSI standards may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

7.1.3 NEMA standards. Copies of NEMA standards may be obtained from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

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7.1.4 NFPA standards. Copies of NFPA standards may be obtained from the National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02269-9101.

7.1.5 OSHA standards. Copies of OSHA standards may be obtained from the U.S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

7.2 Ordering data. Acquisition documents must specify the following information:

a. Title, number, and date of this document.
b. Type, class, style, and size of cutoff machine required (see 2).
c. Workpiece holder, if different (see 3.2.2).
d. Workpiece holder type, as specified (see 3.2.2).
e. Workpiece holder control (style A), as specified (see 3.2.2).
f. Workpiece holder horizontal swivel angle, as specified (see 3.2.2).
g. Wheelhead, if different (see 3.2.3).
h. Wheelhead (class 2) power-drive mechanism control, as specified (see 3.2.3).
i. Wheelhead vertical swivel angle, as specified (see 3.2.3).
j. Spindle drive motor horsepower, if different (see 3.2.3.2).
k. Abrasive disk, if different (see 3.2.3.4).
l. Circular saw blade, as specified (see 3.2.3.5).
m. Flanges, if different (see 3.2.3.6).
n. Oscillating (reciprocating) mechanism, if required (see 3.2.4).
o. Electrical system voltage, as specified (see 3.2.9).
p. Termination of wiring, as specified (see 3.2.9).
q. Spindle drive motor horsepower, if different (see 3.2.10).
r. Motor enclosure, if different (see 3.2.10).
s. Optional accessories, as specified (see 3.2.11).
t. Performance, if different (see 3.3).
u. Machine dimensions and weight, as specified (see 3.4).
v. System of units required (see 3.6).
w. Dual system of units, if required (see 3.6).
x. Lubrication plate or chart, if required (see 3.7).
y. Nameplate, if different (see 3.8).
z. Testing, if different (see 5.3).
aa. Test responsibility, if different (see 5.3.1).
bb. Certification of calibration, as specified (5.3.1).
cc. Optional tests, as specified (see 5.3.7).
dd. Acceptance, if different (see 5.4).
ee. Packaging requirements, as specified (see 6.1).

7.3 Part identification number (PIN). The following part identification numbering system is for government purposes and does not constitute a requirement for the contractor.

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- Size: (see table I for size)
- Style: A - Style A – Floor mounted
  B - Style B – Bench mounted
- Class: 1 - Class 1 – Manual operated wheelhead
  2 - Class 2 – Power operated wheelhead
- Type: A - Type I – Dry cut
  B - Type II – Wet cut
- CID number

MILITARY INTERESTS:

Custodians:
Air Force - 99
Army - AL
Navy - SH

Reviewers:
Air Force - 84
Army - SM

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 6FET

Preparing activity:
DLA - GS

(Project 3405-0101)

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